Maintenance Schedule

Diesel engine-generator sets

DG10V1600A1E
DG10V1600A2E
DG12V1600A1E
DG12V1600A2E
Application group 3F

MS50809/01E

A Rolls-Royce solution

System designation		Engine model	kW/cyl.	Application group
Product designation	Model number			
mtu 10V1600 DS500	DG10V1600A1I	10V1600G10F	40.7 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN
mtu 10V1600 DS540	DG10V1600A2I	10V1600G20F	44.8 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS650	DG12V1600A1I	12V1600G10F	43.7 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS715	DG12V1600A2I	12V1600G20F	48.0 kW/cyl.	3F, Heavy duty for DCP, unrestricted, ICXN

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Special conditions

Designation	Value
Load factor	≤ 75%
Operating hours	Unlimited hours
Time between major overhaul	12,000 h

Table 2: Special conditions

Note 01:
The engines are certified for operation with the fuels approved in the Fluids and Lubricants Specifications. For operation with a high sulfur content in the fuel, the following must be observed: • The component TBO specified in the maintenance schedule refers to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590 or ULSD. If a fuel with a sulfur content > 15 ppm is used, the times specified in the maintenance schedule for component TBO of the cylinder head may be shorter. For reduced TBOs, refer to the Fluids and Lubricants Specifications. Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

Note 02:
If the engine is used in an UPS unit with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions

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1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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